

Faithful Cognitive Representatives

Vassilis Markos¹ and Loizos Michael^{1,2}

¹*Open University of Cyprus, Nicosia, Cyprus*

²*CYENS Center of Excellence, Nicosia, Cyprus*
vasileios.markos@st.ouc.ac.cy, loizos@ouc.ac.cy

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Abstract: In this paper we propose and demonstrate a human–computer–interaction methodology allowing the accurate and faithful extraction of user knowledge. We demonstrate its efficiency in a controlled artificial setting, showcasing how learning in this context is both efficient and faithful to the user’s knowledge and intentions. Finally, we discuss potential areas of application for this machine learning paradigm.

1 Introduction

Artificial Intelligence (AI), and Generative AI (GenAI) in particular, has found a plentiful applications in nearly all aspects of everyday life. On several occasions, the underlying tasks can be formulated as search problems, in the sense of having an AI agent move around a state space, starting from an initial state, crafting a path towards a goal state. Examples of this might include route finding, puzzle solving, shopping planning, code debugging, clinical counseling and so on. Furthermore, restrictions are put on the shape of the *optimal* path, in accordance to the nature of the task at hand, e.g., seeking a shortest path for route finding (Liu, 1996; Migel et al., 2024), finding a particular kind of solution to a puzzle (Mutalova et al., 2023), or strictly following a therapist’s guidance to provide further advice to a client (Sedlakova and Trachsel, 2023; Fiske et al., 2019).

Notably, in all such cases, path optimality can also be understood in the context of conformance to the search policy of a coach, be it explicit, e.g., in a therapeutic setting, or implicit, e.g., in the case of route finding. This kind of formulation allows one to also control under appropriate learning semantics, not only *what* the learner learns to accomplish, but also *how* it does so, which might be of higher significance in particular settings. In this work, we put forward such a search coaching paradigm inspired by previous work (Michael, 2019; Markos and Michael, 2022) in a simplified setting, exploring this interplay between the end and the means to it learned by an agent.

2 Related literature

Knowledge exchange between a coach and a machine learner in the form of context–driven advice giving based on the learner’s decisions has been described as soon as in (McCarthy, 1959). Recently, it has been formalized in the context of *Machine Coaching* (Michael, 2017; Michael, 2019) under the Probably Approximately Correct (PAC) model (Valiant, 1984).

A similar approach, eXplainable Interactive Learning (XIL), (Teso and Kersting, 2019) also considers a human supervisor in the role of a learning “coach”, with advice either correcting mislabeled data, or wrong explanations provided by the learner for otherwise correct predictions. Nevertheless, while in Machine Coaching coach advice can be directly embedded into the learner’s knowledge base in an elaboration–tolerant manner (McCarthy, 1998), XIL utilizes an active learning algorithm as a black–box (Settles, 2009) alongside a local post-hoc explainer (Ribeiro et al., 2016). The corresponding explanations are then used to generate additional labeled data for the learner, to help it cure any misunderstandings.

User feedback can also be utilized to proactively reduce the need for data point labeling, as in (Raghavan and Allan, 2007) for Support Vector Machines. In the same spirit, (Settles, 2011) proposes an interactive framework allowing users to label selected data points with the aim of further polishing the training of a document classifier, while in (Zaidan et al., 2007), a machine learning paradigm utilizing user–provided explanations to reduce learning complexity is proposed. Interestingly, in (Zaidan et al., 2007) it is stressed that an alternative to immense data load

for performance improvement is *data richness*, i.e., the provision of more meaningful data, e.g., in the form of a correct label accompanied by an explanation for it. *Coactive Learning*, proposed in (Shivaswamy and Joachims, 2015), is another interactive Machine Learning framework where a learner receives user advice that is, inherently, only “slightly better” than the learner’s current prediction. The main benefit of the approach, as shown in (Shivaswamy and Joachims, 2015), is that this approach can be integrated into many existing black-box methodologies.

Besides guaranteeing faithfulness and effectively building user trust (Barredo Arrieta et al., 2020), explanation embedding during model training has been reported to improve testing performance (Selvaraju et al., 2019). Moreover, utilizing bilateral explanations, i.e., from the learner towards the coach and vice versa, can further speed up the learning process (Dong et al., 2017), with similar results also obtained in model retraining upon the provision of additional explanations regarding both wrong and correct, but for the wrong reasons, model decisions (Rieger et al., 2019). At last, post-hoc model fine tuning also leads to performance improvements (Ribeiro et al., 2016).

Given the recent spike in the use of Large Language Models (LLMs) in virtually all domains, human-machine interaction methodologies that utilize human expertise to further polish a LLM have been proposed. In (Welz and Lanquillon, 2024) a human knowledge infusion methodology is presented, where expert users provide domain specific knowledge that is used to populate a knowledge base, then utilized by the system in tandem with the LLM to generate responses to user queries.

3 Sorting coaching

In this Section we empirically explore search coaching, by having a machine learner iteratively seek advice by a coach, as in (Markos et al., 2022), with the aim of iteratively improving its search capacity, guided by the coach’s explicit search policy. Given a fixed goal state, g , the learner applies its search policy, $\mathbb{P}_{\text{learner}}$, to identify a path, p , starting from an arbitrary state, s , and ending at a goal state, g . Upon lack of knowledge, the learner seeks advice from its coach, which responds according to its own policy, $\mathbb{P}_{\text{coach}}$. Each piece of advice has the form of an “if-then” rule, where the “if” part is state-dependent and the “then” part is always an action to be undertaken by the learner. Rules are prioritized by the coach as in previous works (Markos et al., 2022; Markos and Michael, 2022; Ioannou and Michael, 2021) with the

last received ones overriding previous knowledge. A *coaching step* is a single learner request for advice and the corresponding coach response, while a *coaching session* comprises of the full learning cycle, where the learner eventually crafts one such path or the process is aborted (cf Algorithm 1).

Data: $s, g, \text{learner}, \text{coach}$

Result: *learner* with updated search policy.

$\text{advice} \leftarrow \text{null};$

do

$\text{learner.update_hypothesis}(\text{advice});$

$\text{path} \leftarrow \text{learner.search_path}(s, g);$

$\text{advice} \leftarrow \text{coach.evaluate}(\text{path}, s, g);$

while advice is not null;

Algorithm 1: Search coaching process. s, g denote the start and goal states, respectively.

3.1 Experimental setup

We have implemented two machine coaches, $\mathcal{C}_{\text{bubble}}$, which uses bubble sort as its underlying coaching policy, and $\mathcal{C}_{\text{quick}}$, which uses quicksort for that purpose, both resulting to advice corresponding to the next step in the algorithm underlying each coaching policy. A state of size n in this case is any permutation of $\{1, 2, \dots, n\}$, with start states being arbitrarily chosen among all possible states of the same size and the goal state is always $(1, 2, \dots, n)$. Rules in this case have a potentially partial permutation of $\{1, 2, \dots, n\}$ as their condition and a swap action between two items of this permutation, as shown in Figure 1. We let n range from 1 to 20, running $m = 100$ iterations for each value of n . We have also explored two different types of advice for each coach: (A1) *full advice*, where each piece of advice depends on the entire state, and; (A2) *partial advice*, where each piece of advice depends on the specific part of the state to be improved. Moreover, for each advice type we have considered three learner configurations regarding advice memory: (M1) *no memory* across different iterations for the same state size, n , i.e., coaching sessions are pairwise independent; (M2) *short memory* across different iterations for the same state size, n , i.e., coaching sessions across different values of n are independent but not within the same value for n , and; (M3) *long memory* across all values of state size, n , so all coaching sessions are correlated, building on knowledge from previous test cases. In all cases the learner seeks advice upon lack of knowledge for further searching, returning a partial path p starting from s , while the coach adopts one of the following *coaching styles* as illustrated in Figure 2: (C1) *reactive* coaching, where

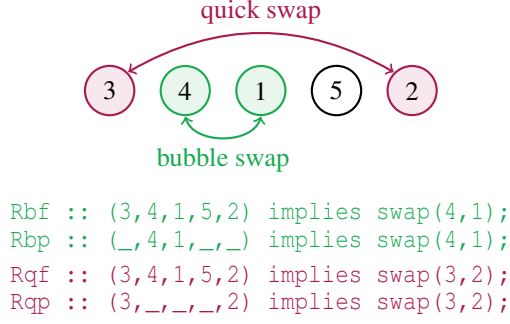


Figure 1: Examples of full (f) and partial (p) advice from C_{bubble} (b) and C_{quick} (q) on $(3, 4, 1, 5, 2)$.

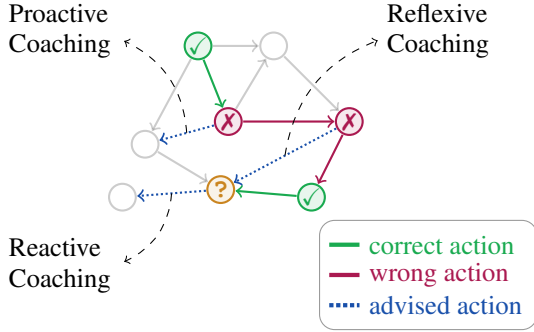


Figure 2: Different coaching styles used in our experimental setup. Upon the learner reaching a state from where there is no knowledge on how to proceed (?), a reactive piece of advice resolves this lack of knowledge, a proactive piece of advice addresses the first wrong decision in the learner’s search path, while a reflexive piece of advice addresses a random mistake / lack of knowledge of the learner.

coach provides advice on learner’s last wrong decision in p ; (C2) *proactive* coaching, where coach provides advice on learner’s earliest wrong decision in p , and; (C3) *reflexive* coaching, where coach provides advice at a random wrong decision in p .

3.2 Results

In Figures 3 we present the number of coaching interactions required to reach the goal state in each case as a function of the state size, n , for all three types of coaching. Considering reactive coaching and full advice (Fig. 3a), the utilization of either short or long memory does not appear to have any particular effect on the number of coaching steps, which was expected given the restricted applicability of provided advice. On the contrary, partial advice (Fig. 3d) combined with both short and long memory results to fewer coaching steps, which can be symmetrically explained by the broader applicability of the provided advice. Also observe that in the case of full advice, coaching steps are proportional to the big-O com-

plexity of the underlying policy / sorting algorithm, which, again, comes as a result of limited advice applicability, forcing the learner to essentially ask for step-by-step guidance in each separate case. In juxtaposition to the full advice case, utilizing memory significantly reduces the required coaching steps per session, since now the learner can effectively apply previously acquired knowledge to new cases. Furthermore, note that in the long memory case (Fig. 3d), coaching steps for both C_{bubble} and C_{quick} are essentially the same (dotted lines).

Considering proactive coaching, where the learner receives advice on its first wrong decision (Figs. 3b, 3e), we observe that full advice imposes the same trends in coaching step growth as with reactive coaching. However, providing partial advice, leads to a significantly reduced number of coaching steps per session, as with reactive coaching, since, utilizing memory, the learner is able to generalize from previous cases. Notably, coaching steps are systematically higher from the corresponding reactive coaching cases (Figs. 3d, 3e). This reflects the more “meticulous” way a proactive coach interacts with the learner, providing (even partial) advice that is more accurate, preventing any unnecessary exploration of the search space, at the cost of having to interact more frequently with the learner, compared to a more “lazy” reactive coach, which prefers to just react to a learner’s demand for further knowledge.

Regarding reflexive coaching (Figs. 3c, 3f), proactive coaching trends are still observed, in the same fashion, on average. However, in this case, the coach, randomly selecting a state to intervene, might provide the same piece of advice more than once. We report the number of such sessions as *trivial runs*. As state size grows larger, so does the number of trivial runs, especially in short / long memory cases, where the learner has a, typically, larger policy, hence it is more probably to be exposed to the same piece of advice more than once.

Finally, we observe that coach type is irrelevant in case the learner is equipped with some kind of memory, regardless of advice type. For full advice this is expected given the restricted advice applicability which prohibits the learner from reusing previous knowledge, while partial advice, on the contrary, allows the learner to generalize conforming to P_{coach} , reducing the need for coach feedback. Focusing on partial advice in the absence of memory, reactive coaching results to coaching sessions of size proportional to the coaching policy complexity, as noted above. However, for proactive and reflexive coaching this is not true, as the coach appears to interact with the learner significantly more frequently, indi-

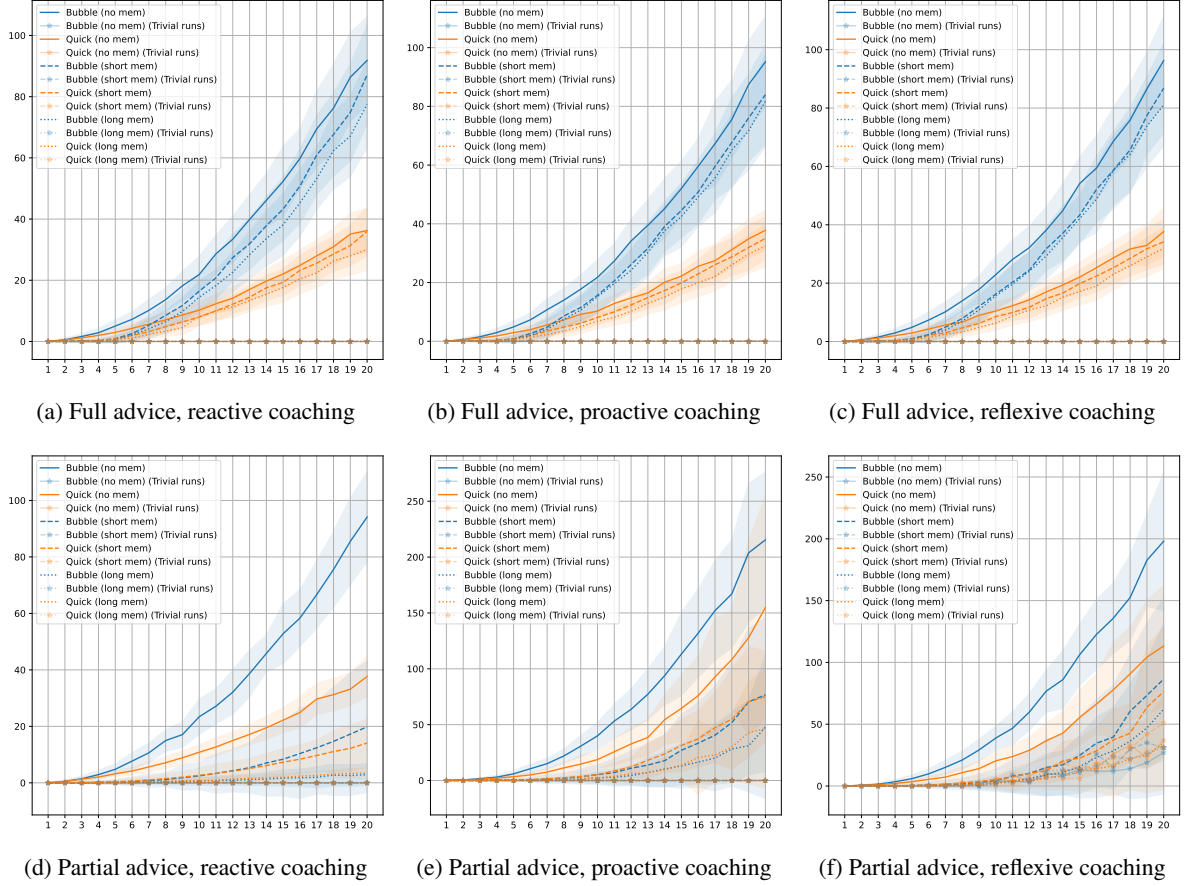


Figure 3: Coaching steps as a function of state size, n . Full and partial advice (top and bottom, respectively) with reactive, proactive and reflexive coaching styles are considered (left, middle, and right, respectively), with all possible coach (C_{bubble} , C_{quick}) and memory (none, short, long) combinations are shown.

cating that in the reactive case the learner might end up learning an efficacious but unfaithful policy, which might not be true for proactive / reflexive coaching.

4 Coaching fidelity

Motivated by the potential learner unfaithfulness discussed before, we explore how much the learner conforms to the coach’s instructions in the various settings explored in Section 3. To that end, we define *coaching fidelity* to be the relative frequency of learner decisions conforming to coach decisions on the same states over all such decisions in a session.

4.1 Experimental setup

First, we sample $m = 20$ repetitions for state size $n \in \{5, 10, 15, 20\}$ and compute coaching fidelity for each one, across all coach types, coaching styles,

and memory configurations. Then, aiming to shed more light on the parameters affecting coaching fidelity, we introduce a new type of advice, *partial@k*, $2 \leq k \leq n$, where $k - 2$ randomly selected state variables are included into the advice condition, alongside the two required to describe the swap action of interest. Such advice captures intermediate levels of specificity between the two extremes explored so far, with *partial@2* corresponding to partial and *partial@n* to full advice for state size n . We also define *advice coverage* as the percentage of state variables that is included in the condition of a piece of advice. So, e.g., a rule like $R :: (_, 2, 1, _, 5)$ implies $\text{swap}(2, 5)$ has an advice coverage of $3/5 = 60\%$. Again, we ran $m = 20$ iterations for state sizes $n \in \{5, 10, 15, 20\}$ and $2 \leq k \leq n$ in each case, for all coach types, coaching styles, and memory configurations.

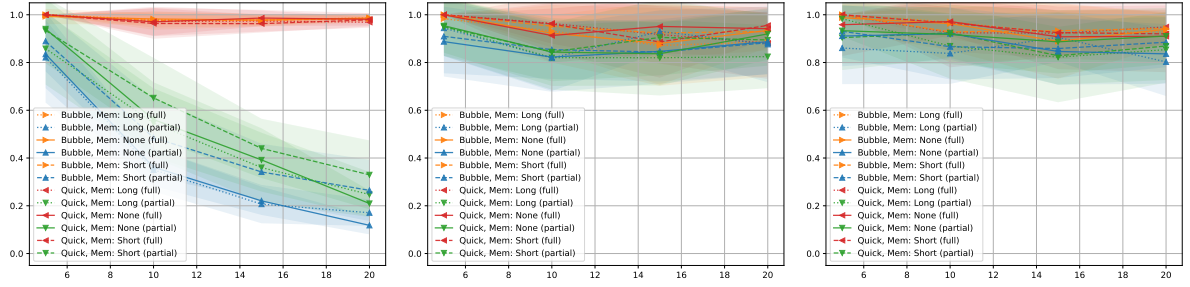


Figure 4: Coaching fidelity against state size for reactive (left), proactive (middle) and reflexive (right) coaching.

4.2 Results

In Figure 4 we show average coaching fidelity as a function of state size, n . Regardless of memory configuration, full advice combined with reactive coaching result to a virtually fully conformant learner, replicating the coach’s policy. Similar results are also obtained for proactive and reactive coaches. However, with partial advice, coaching fidelity depends on the coaching style and the corresponding memory configuration. In particular, in the case of reactive coaching, partial advice results to virtually vanishing fidelity for all memory configurations. For proactive coaching, however, partial advice results to a constantly high fidelity for all examined state sizes, also barely affected by the corresponding memory configuration. Similarly, reflexive coaching also results to higher fidelity than reactive coaching, roughly replicating results for proactive coaching. Also, observe how in all cases coach type does not affect fidelity, except for partial advice reactive coaching, where any differences appear to vanish as state size grows.

Overall, Figure 4 hints towards a trade-off between advice specificity and coaching fidelity. The more specific the advice the more conformant the learner to the coaching policy and, conversely, the more loose the advice, the more the learner generalizes utilizing previous knowledge.

In Figure 5 we present coaching fidelity against advice coverage for all coach types and coaching styles, aggregating on state size. Observe, at first, how coach type and memory configuration do not affect coaching fidelity, while the most significant effect comes from different coaching styles. Indeed, in the case of reactive coaching, lower advice coverage results to the learner eventually reaching the goal state but in a way not conforming to the coach’s search policy, while the more specific the provided advice, the more accurate the learner in its searching steps towards the goal state. On the contrary, for proactive and reflexive coaching, coaching fidelity is just slightly increasing with respect to advice coverage,

being relatively high, even for lower values of advice coverage. Compared to reactive coaching, proactive and reflexive coaching appear to guarantee a constant high fidelity, mitigating the negative effects of arbitrarily low coverage almost equally well. In a sense, intervening early on guarantees that the learner cannot deviate much from the coach’s guidance, even in case where advice is rather minimal.

5 Discussion and future work

In the above, we have explored a machine learning paradigm that allows for the efficacious and, under certain configurations, faithful transfer of knowledge from a coach to a machine learner. A limitation of our design is that the list sorting task is highly structured, admitting a single solution, which might not match many real-world scenarios. Nevertheless, this stiffness makes the sorting task ideal for a proof of concept study design. Moreover, the presented in depth analysis in terms of different coach types, coaching styles and learner memory types effectively mitigates this restriction in the context of exploring and discussing the key features of the proposed learning paradigm. Another limitation stems from the used knowledge representation, which has been artificially restricted to linearly ordered propositional knowledge bases, with rules of a task-specific form. However, utilizing richer representations, as in other works (Michael, 2019; Markos and Michael, 2022; Ioannou and Michael, 2021) would make policy description trivial, since, e.g., bubble sort can be easily described by a single relational rule. This, in turn, would result to single-step coaching sessions, regardless of coach type, coaching style, or memory configuration, making any substantial analysis in this setting impossible or trivial. Hence, restricting to a highly structured yet rich problem with “expressive enough” semantics allows for an in depth initial exploration of the search coaching learning paradigm while does not deduce much value from generalizations that can be

This seems a bit weak, maybe revise later.

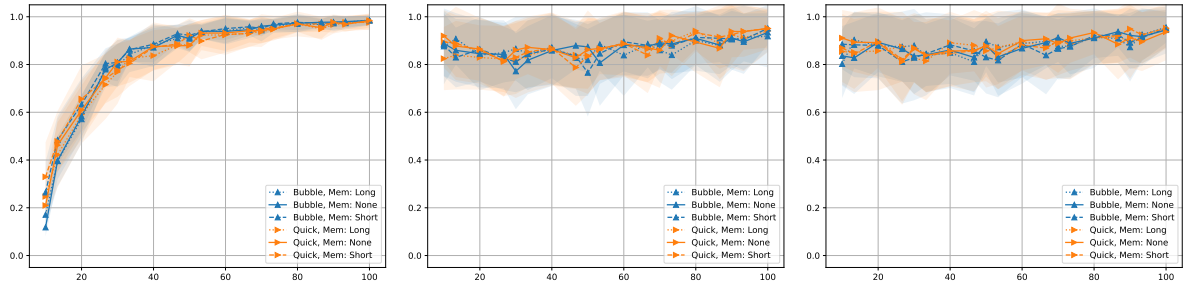


Figure 5: Coaching fidelity against advice coverage for reactive (left), proactive (middle) and reflexive (right) coaching.

made to broader scopes of application.

A key outcome appears to be that properly shaping an agent’s learning environment strongly affects both the acquired knowledge and the way it is applied, allowing for a wide variety of agent behaviors, aligned with the subtleties of the actual task at hand. For instance, if just reaching a certain target state is the ultimate goal, as, e.g., in naive route finding, then using a lazy reactive coaching style allows for learning such a policy in a minimal number of coaching steps. Furthermore, the utilization of memory by the learner appears to essentially wipe out any coaching policy differences, at least in the case explored in this work. This is a useful feature in the case of coaches with potentially imperfect policies, or merging knowledge from multiple resources / coaches, at the cost of not having any guarantees on *how* the agent will end up solving the task.

On the other hand, in cases where not only *what* the goal state is, but also *how* it is reached matters, the proposed search coaching paradigm can be properly configured in plentiful ways to allow for such a more meticulous learning. As shown, one way this can be achieved is by providing overspecific advice, for virtually any coach type, coaching style, and / or memory configuration, essentially having the coach guide the learner step-by-step. While implying higher and more time consuming coach involvement, this kind of coaching and its independence of any other examined parameters except for advice shape might prove useful in the design of knowledge acquisition methodologies where there are no guarantees on the coaching policy efficiency, coaching style, or learner memory availability, e.g., when acquiring potentially conflicting knowledge from multiple semi-expert users. By utilizing proactive, or even reactive, coaching, at least in uniform probability setting explored here, one might coach an eventually efficacious and faithful learner, with the additional advantage of utilizing just partial advice, if needed. This, again, might correspond to real-world scenarios where users are expert or more careful in the way advice is provided and the

provision of full advice is impossible due to combinatorial or other restrictions.

Overall, >>> Add some future directions here, maybe hint towards real-world applications as in Othello, a clinical setting etc

REFERENCES

- Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Benetot, A., Tabik, S., Barbado, A., Garcia, S., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R., and Herrera, F. (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58:82–115.
- Dong, Y., Su, H., Zhu, J., and Zhang, B. (2017). Improving Interpretability of Deep Neural Networks with Semantic Information.
- Fiske, A., Henningsen, P., and Buyx, A. (2019). Your robot therapist will see you now: Ethical implications of embodied artificial intelligence in psychiatry, psychology, and psychotherapy. *J Med Internet Res*, 21(5):e13216.
- Ioannou, C. and Michael, L. (2021). Knowledge-Based Translation of Natural Language into Symbolic Form. In *Proceedings of the 7th Linguistic and Cognitive Approaches To Dialog Agents Workshop - LaCATODA 2021*, pages 24–32, Montreal, Canada.
- Liu, B. (1996). Intelligent route finding: combining knowledge, cases and an efficient search algorithm’. In *ECAI*, volume 96, pages 380–384. PITMAN.
- Markos, V. and Michael, L. (2022). Prudens: An argumentation-based language for cognitive assistants. In Governatori, G. and Turhan, A.-Y., editors, *Rules and Reasoning*, pages 296–304, Cham. Springer International Publishing.
- Markos, V. T., Thoma, M., and Michael, L. (2022). Machine coaching with proxy coaches. In Kuhlmann, I., Mumford, J., and Sarkadi, S., editors, *Proceedings of the 1st Workshop on Argumentation & Machine Learning co-located with 9th International Conference on Computational Models of Argument (COMMA 2022)*, Cardiff, Wales, September 13th, 2022, volume 3208 of *CEUR Workshop Proceedings*, pages 45–64. CEUR-WS.org.

- McCarthy, J. (1959). Programs with Common Sense. In *Proceedings of the Teddington Conference on the Mechanization of Thought Processes*, pages 75–91, London, U.K.
- McCarthy, J. (1998). Elaboration Tolerance. In *Common Sense 98*, London, U.K.
- Michael, L. (2017). The Advice Taker 2.0. In *Proceedings of the 13th International Symposium on Commonsense Reasoning*, volume 2052, London, U.K.
- Michael, L. (2019). Machine Coaching. In *IJCAI 2019 Workshop on Explainable Artificial Intelligence*, pages 80–86, Macau, China.
- Migel, S., Maloied, M., Zaliskyi, M., Lelechenko, A., Osipchuk, A., and Solomentsev, O. (2024). Optimal pathfinding based on artificial intelligence tools. In *International Workshop on Advances in Civil Aviation Systems Development*, pages 242–261. Springer.
- Mutalova, R., Sarrazin-Gendron, R., Ghasemloo Gheidari, P., Cai, E., Richard, G., Caisse, S., Knight, R., Blanchette, M., Szantner, A., and Waldispühl, J. (2023). Player-guided ai outperforms standard ai in sequence alignment puzzles. In *Proceedings of The ACM Collective Intelligence Conference, CI '23*, page 53–62, New York, NY, USA. Association for Computing Machinery.
- Raghavan, H. and Allan, J. (2007). An Interactive Algorithm for Asking and Incorporating Feature Feedback into Support Vector Machines. In *Proceedings of the 30th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR '07*, pages 79–86, New York, NY, USA. Association for Computing Machinery.
- Ribeiro, M. T., Singh, S., and Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '16*, pages 1135–1144, New York, NY, USA. Association for Computing Machinery.
- Rieger, L., Singh, C., Murdoch, W. J., and Yu, B. (2019). Interpretations are Useful: Penalizing Explanations to Align Neural Networks With Prior Knowledge.
- Sedlakova, J. and Trachsel, M. (2023). Conversational artificial intelligence in psychotherapy: A new therapeutic tool or agent? *The American Journal of Bioethics*, 23(5):4–13.
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2019). Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *International Journal of Computer Vision*, 128(2):336–359.
- Settles, B. (2009). Active Learning Literature Survey. Technical Report, University of Wisconsin-Madison Department of Computer Sciences.
- Settles, B. (2011). Closing the Loop: Fast, Interactive Semi-Supervised Annotation With Queries on Features and Instances. In *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing*, pages 1467–1478, Edinburgh, Scotland, UK. Association for Computational Linguistics.
- Shivaswamy, P. and Joachims, T. (2015). Coactive Learning. *Journal of Artificial Intelligence Research*, 53:1–40.
- Teso, S. and Kersting, K. (2019). Explanatory Interactive Machine Learning. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society, AIES '19*, pages 239–245, New York, NY, USA. Association for Computing Machinery.
- Valiant, L. G. (1984). A Theory of the Learnable. *Communications of the ACM*, 27(11):1134–1142.
- Welz, L. and Lanquillon, C. (2024). Enhancing large language models through external domain knowledge. In Degen, H. and Ntoa, S., editors, *Artificial Intelligence in HCI*, pages 135–146, Cham. Springer Nature Switzerland.
- Zaidan, O., Eisner, J., and Piatko, C. (2007). Using "Annotator Rationales" to Improve Machine Learning for Text Categorization. In *Human Language Technologies 2007: The Conference of the North American Chapter of the Association for Computational Linguistics; Proceedings of the Main Conference*, pages 260–267, Rochester, New York. Association for Computational Linguistics.